Roll No. | | | | | | | |

End Semester Examination 2025

Name of the Course: ***M. Tech (Electric Vehicle Technology)*** Semester: 1st
Name of the Paper – **Research Methodology & IPR**
Paper Code: MEV 106

Time: 3Hours Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Describe the approaches for investigating solutions to a research problem. Explain in detail the process of data collection, analysis, interpretation, and the necessary instrumentation used for conducting reliable and valid research.	
(b)	Describe the various sources of research problems. How can factors such as literature surveys, industrial requirements, environmental challenges, and theoretical gaps lead to the identification of a new research problem?	**CO1**
(c)	Discuss the criteria and characteristics of a good research problem. Explain the common errors made while selecting a research problem, and suggest methods to overcome them.	
Q2		
(a)	Describe the steps involved in developing a research proposal. What are the essential components that must be included in a proposal to make it complete and acceptable for review?	
(b)	Discuss the key elements of effective technical writing. Explain the structure, clarity, and presentation required while writing a research report or thesis.	**CO2**
(c)	What is plagiarism? Discuss its different types and explain how it can be avoided. Also highlight the importance of research ethics in maintaining the quality and credibility of scientific research.	
Q3		**CO3**
(a)	Explain the application of measures of central tendency — Mean, Median, Mode, Geometric Mean, and Harmonic Mean — in quantitative data analysis. Discuss with suitable examples from engineering or scientific research.	
(b)	Describe the concept of correlation analysis. Explain the types of correlation and methods used to determine the degree and direction of relationship between two variables.	
(c)	Explain the application of ANOVA (Analysis of Variance) and ANCOVA (Analysis of Covariance) in research data analysis. Discuss their importance in testing the significance of experimental results.	
Q4		**CO4**
(a)	Explain the nature and importance of Intellectual Property Rights (IPR). Discuss in detail the different types of intellectual property — Patents, Industrial Designs, Trademarks, and Copyrights — along with suitable examples.	

(b)	Describe the process of patenting and technological development. Explain how technological research and innovation lead to the creation of patentable inventions, and outline the steps involved in filing a patent application.	
(c)	Explain in detail the procedure for the grant of a patent in India. Discuss the major steps involved from application ,filing to publication, examination, opposition, and final grant of the patent.	
Q5		**CO5, CO6**
(a)	Describe the process of licensing and transfer of technology. How do patents facilitate technology transfer and industrial collaboration? Provide examples to illustrate your answer.	
(b)	Discuss the new developments in Intellectual Property Rights (IPR) with reference to the administration of the patent system, IPR protection for biological systems and computer software, and traditional knowledge. Support your answer with suitable case studies.	
(c)	Define Geographical Indications (GI) and explain their role in protecting local products and traditional knowledge. Give examples of Indian products registered under the GI system.	